Realised Volatility Forecasting — Multi-Asset Benchmark

Direct Multi-Step Models: HAR, NLinear, DLinear, SAS, RNN

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1 Setup

Input series. Either raw $y_t = \mathrm{RV}_t > 0$ (default) or log-transformed $y_t = \log \mathrm{RV}_t \in \mathbb{R}$ (`--log`). All internal scaling (z-scoring) is handled inside each model class.

Forecasting task. Direct multi-step: one readout per $h \in \{1, 5, 10\}$ days. Readouts are ridge-regression with $\alpha \in \{10^k : k \in [-4, 5],\ \Delta k{=}0.25\}$ (37 values), 5-fold time-series CV.

2 Models

2.1 HAR — Heterogeneous Autoregressive Model (Corsi 2009)

Features. $\bar{y}_t^{(w)} = \frac{1}{5}\sum_{k=0}^{4} y_{t-k}, \quad \bar{y}_t^{(m)} = \frac{1}{22}\sum_{k=0}^{21} y_{t-k}$.

$$\hat{y}_{t+h} = \alpha + \beta_d\, y_t + \beta_w\, \bar{y}_t^{(w)} + \beta_m\, \bar{y}_t^{(m)} \tag{1}$$

HAR-OLS: $\min_\beta \|X\beta - y\|_2^2$. **HAR-Ridge**: column-normalise features; penalise slopes only; back-transform coefficients after fit.

2.2 NLinear and DLinear (Zeng et al. 2023)

Lookback window $\boldsymbol{y}_t = (y_{t-L+1}, \ldots, y_t)^\top \in \mathbb{R}^L$.

NLinear — subtract last value (removes level, keeps increments):

$$\tilde{\boldsymbol{y}}_t = \boldsymbol{y}_t - y_t, \qquad \hat{y}_{t+h} = \tilde{\boldsymbol{y}}_t^\top w + y_t \tag{2}$$

DLinear — trend/seasonality decomposition:

$$\boldsymbol{\tau}_t = \mathrm{MA}_k(\boldsymbol{y}_t), \quad \boldsymbol{\varsigma}_t = \boldsymbol{y}_t - \boldsymbol{\tau}_t, \qquad \hat{y}_{t+h} = \begin{bmatrix} \boldsymbol{\tau}_t \\ \boldsymbol{\varsigma}_t \end{bmatrix}^\top w \tag{3}$$

2.3 GARCH — AR(p)-GARCH(1,1)

$y_t = \mu + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t, \quad \sigma_t^2 = \omega + \alpha_G \varepsilon_{t-1}^2 + \beta_G \sigma_{t-1}^2$. Multi-step: iterate variance recursion analytically. Estimated by QMLE.

2.4 SAS — State Affine System

Diagonal basis. $P_0, \ldots, P_p \in \mathbb{R}^n,\ Q_0, \ldots, Q_q \in \mathbb{R}^n$, input $z \in \mathbb{R}$.

$$P(z) = \mathrm{clip}\Big(\sum_{d=0}^{p} P_d\, z^d,\ -1{+}\epsilon,\ 1{-}\epsilon\Big) \in (-1,1)^n \tag{4}$$

$$Q(z) = \sum_{d=0}^{q} Q_d\, z^d \in \mathbb{R}^n \tag{5}$$

$$s_t = P(z_t) \odot s_{t-1} + Q(z_t) \tag{6}$$

Clipping guarantees $|P(z)_i| < 1$ (unconditional stability).

Standard readout.

$$\hat{y}_{t+h} = s_t^\top w_h, \qquad w_h = \arg\min_w \|S\,w - Y_h\|^2 + \alpha\|w\|^2 \tag{7}$$

where $S \in \mathbb{R}^{(T-p-h)\times n}$ are post-washout states and $Y_h \in \mathbb{R}^{T-p-h}$ are z-scored targets y_{t+h}.

Residual-target variant ($\mathbf{SAS_{res}}$). Instead of fitting the readout on the level y_{t+h}, fit it on the increment $r_{t,h} = y_{t+h} - y_t$ (per-horizon z-scored separately):

$$w_h = \arg\min_w \big\|S\,w - \tilde{r}_h\big\|^2 + \alpha\|w\|^2, \qquad \tilde{r}_h = \frac{r_h - \mu_{r,h}}{\sigma_{r,h}} \tag{8}$$

At prediction time reconstruct the level using the anchor y_t:

$$\hat{r}_{t,h} = \sigma_{r,h}\,(s_t^\top w_h) + \mu_{r,h}, \qquad \hat{y}_{t+h} = y_t + \hat{r}_{t,h} \tag{9}$$

The anchor $y_t > 0$ prevents near-zero predictions and stabilises QLIKE. Scalers $(\mu_{r,h}, \sigma_{r,h})$ are computed per horizon on the training window and updated at each refit.

Initialisation of P. Let ρ be the spectral-norm bound and $\delta = \max(0.005,\ (1{-}\rho)\cdot 0.3)$.

- $P_0[i] \overset{\text{iid}}{\sim} \mathcal{U}(-\rho,\ \rho)$
- $P_k[i] \overset{\text{iid}}{\sim} \mathcal{N}(0,\ \delta/5^k)$ for $k \geq 1$

Initialisation of Q. To keep injection variance uniform across degrees (for $z \sim \mathcal{N}(0,1)$, $\mathbb{E}[z^{2k}] = (2k-1)!!$):

$$Q_k[i] \overset{\text{iid}}{\sim} \mathcal{N}\left(0,\ \frac{1}{\sqrt{n\,(2k-1)!!}}\right) \tag{10}$$

Associative structure. Composition rule on pairs $\ell_t = (P(z_t), Q(z_t))$:

$$(A_j,\, b_j) \circ (A_i,\, b_i) = \big(A_j \odot A_i,\ \ A_j \odot b_i + b_j\big) \tag{11}$$

This is associative $\Rightarrow$ parallel prefix scan (depth $O(\log T)$ vs. $O(T)$ sequential).

Algorithm 1: Two-Level Parallel Scan — compute $s_1, \ldots, s_T$

Input : Transitions $\ell_1, \ldots, \ell_T$;;
chunk size B;;
$s_0 = \mathbf{0}$
Output : States $s_1, \ldots, s_T$

$K \leftarrow \lceil T/B \rceil$; pad to KB
Phase 1 (parallel over K chunks): $(A^{\text{intra}}_{k,\cdot}, b^{\text{intra}}_{k,\cdot}) \leftarrow \textsc{AssocScan}(\circ,\ \ell_{kB+1}, \ldots, \ell_{(k+1)B})$
Phase 2 (inter-chunk scan over K endpoints):
$(A^{\text{inter}}, b^{\text{inter}}) \leftarrow \textsc{AssocScan}(\circ,\ (A^{\text{intra}}_{1,B}, b^{\text{intra}}_{1,B}), \ldots)$
Phase 3 (carry): $c_0 = s_0$;;
$c_k = A^{\text{inter}}_{k-1} \odot s_0 + b^{\text{inter}}_{k-1}$
Phase 4 (parallel over K chunks): $s_{kB+j} = A^{\text{intra}}_{k,j} \odot c_k + b^{\text{intra}}_{k,j}$

2.5 RNN — LSTM / GRU / LRU

Readout: $\hat{y}_{t+h} = \text{Linear}_h(h_T)$, one head per h. Training: Adam, early stopping (patience = 20, val = 20% chronological).

LRU (Orvieto et al. 2023):

$$h_t = \sigma(\nu) \odot h_{t-1} + Bz_t, \qquad y_t = Ch_t, \qquad \sigma(\nu) \in (0,1)^d \tag{12}$$

3 Evaluation Protocol

3.1 Rolling Out-of-Sample Loop

Each model owns its preprocessing (z-scoring, log-transform). The loop passes raw series values unchanged.

Algorithm 2: Rolling OOS Evaluation

Input : Series $y_1, \ldots, y_T$; window W; refit freq R; horizons $\mathcal{H}$
Output : Records $\{(\hat{y}_{t+h,m},\ y_{t+h},\ \text{sqerr},\ \text{qlike})\}$

for $t = W$ **to** $T - \max(\mathcal{H})$ **do**
  if *steps since last refit* $\geq R$ **then**
    foreach *model* m **do**
      $m.\textsc{Fit}(y_{t-W:t},\ \mathcal{H})$
    ; `// model z-scores internally`
  foreach *model* m **do**
    $m.\textsc{Update}(y_t)$
  ; `// advance state to` t
  foreach *model* m, $h \in \mathcal{H}$ **do**
    $\hat{y}_{t+h} \leftarrow m.\textsc{Predict}(h)$; `// in original scale`
    record $\big(\hat{y}_{t+h},\ y_{t+h},\ (\hat{y}_{t+h} - y_{t+h})^2,\ \mathcal{L}(\hat{y}_{t+h}, y_{t+h})\big)$;

3.2 Loss Functions

Mean Squared Error.

$$\text{MSE}^{(h)}_m = \frac{1}{T_{\text{oos}}} \sum_{t=W}^{T-h} \big(\hat{y}_{t+h,m} - y_{t+h}\big)^2 \tag{13}$$

QLIKE (Patton 2011) — level-space ($y_t > 0$, default):

$$\mathcal{L}(y, \hat{y}) = \frac{y}{\hat{y}} - \log \frac{y}{\hat{y}} - 1 \qquad (\hat{y} \text{ clipped to } 10^{-8}) \tag{14}$$

$\mathcal{L} \geq 0$ with minimum 0 at $\hat{y} = y$. Scale-free: $\mathcal{L}(ky, k\hat{y}) = \mathcal{L}(y, \hat{y})$. Strongly penalises underforecasting ($\hat{y} \ll y$).

QLIKE — log-space (`--log` mode, $y_t \in \mathbb{R}$): Apply the level formula to e^{y_t}, $e^{\hat{y}_t}$:

$$\mathcal{L}_{\log}(y, \hat{y}) = \frac{e^{\hat{y}}}{e^{y}} - (y - \hat{y}) - 1 = e^{\hat{y}-y} + (\hat{y} - y) - 1 \tag{15}$$

Valid for any finite $y, \hat{y}$ (no positivity constraint required).

3.3 Model Confidence Set (Hansen et al. 2011)

Given loss matrix $L \in \mathbb{R}^{T \times M}$ (raw squared errors, one column per model):

1. $d_{t,ij} = L_{t,i} - L_{t,j}$; $\quad \bar{d}_{ij} = T^{-1} \sum_t d_{t,ij}$
2. $T_R = \max_{m \neq b} \bar{d}_{mb} / \hat{\sigma}_{mb} \quad$ ($\hat{\sigma}^2_{mb}$: circular block bootstrap, B reps, block width $w = \lfloor T^{1/3} \rfloor$)
3. Eliminate the worst model if p-value$(T_R) < \alpha$; repeat.
4. MCS_α = surviving set.

MCS Frequency. Run MCS independently per symbol per horizon; report survival rate $c_m^{(h)} / N_{\text{sym}}$.

3.4 Diebold-Mariano Test (Harvey-Leybourne-Newbold 1997)

For each (model, horizon, symbol), test whether a challenger beats the benchmark. **Raw loss differences** are used for both MSE and QLIKE:

$$d_t = \mathcal{L}_{t,\text{challenger}} - \mathcal{L}_{t,\text{benchmark}} \tag{16}$$

$H_0 \colon \mathbb{E}[d_t] = 0$; $H_1^< \colon \mathbb{E}[d_t] < 0$ (challenger better).

HAC variance with Newey-West, $h - 1$ lags (MA$(h-1)$ structure of h-step errors):

$$\hat{V} = \hat{\gamma}_0 + 2 \sum_{k=1}^{h-1} \left(1 - \frac{k}{h}\right) \hat{\gamma}_k, \qquad \hat{\gamma}_k = T^{-1} \sum_t (d_t - \bar{d})(d_{t-k} - \bar{d}) \tag{17}$$

HLN finite-sample correction:

$$\mathrm{DM}^* = \sqrt{\frac{T + 1 - 2h + h(h-1)/T}{T}} \cdot \frac{\bar{d}}{\sqrt{\hat{V}/T}} \sim t_{T-1} \tag{18}$$

Report: fraction of symbols where $p^* < \alpha$ (one-sided), per horizon.

4 Hyperparameters

Table 1: Fixed hyperparameters used in all experiments.

Component	**Parameter**	**Value**
Evaluation	Training window W	2 000
	Refit frequency R	20 steps ($\approx$1 month)
	Horizons $\mathcal{H}$	$\{1, 5, 10\}$ days
MCS	Significance α	0.10
	Bootstrap replications B	1 000
HAR	Lags (d, w, m)	$(1, 5, 22)$ days
	Ridge CV folds	5
NLinear / DLinear	Lookback L	20
	DLinear MA kernel	5
GARCH	AR lags p	5
	Estimation	QMLE
SAS (diagonal)	Reservoir size n	200
	Spectral norm ρ	0.95
	P-polynomial degree p	1
	Q-polynomial degree q	1 ($\mathrm{SAS_{q1}}$) or 2 ($\mathrm{SAS_{q2}}$)
	Washout	50 steps
	Residual-target	off (SAS) / on ($\mathrm{SAS_{res}}$)
RNN (GRU)	Hidden size	32
	Layers	1
	Lookback	20
	Epochs (max)	500
	Early stopping	patience = 20, val = 20%
Ridge grid	Range	10^k, $k \in [-4, 5]$
	Spacing	$\Delta k = 0.25$, 37 values
	CV folds	5

5 Results